Point-Based Mask Refinement for X-Ray Instance Segmentation

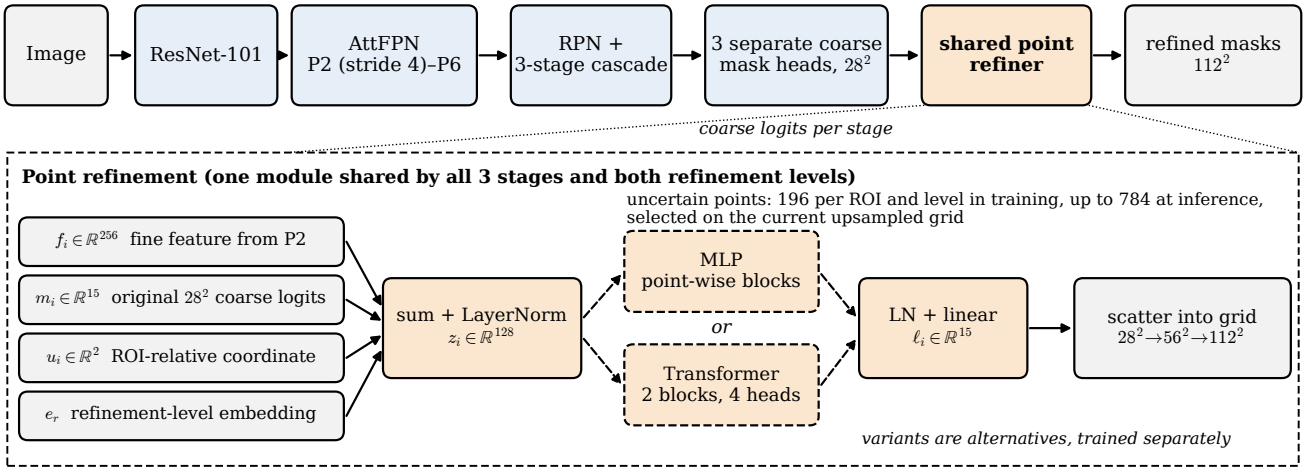


Figure 1: Point-based mask refinement. Three cascade mask heads give coarse 28^2 masks; one shared refiner updates uncertain points at 56^2 and 112^2. MLP and transformer are alternative variants.

1 Method

Region-based detectors predict X-ray instance masks on a coarse 28×28 grid, which limits accuracy for thin and overlapping prohibited items. We refine these masks at selected points (Fig. 1). The parent detector is a three-stage Cascade Mask R-CNN with ResNet-101 and AttFPN; P2 denotes the stride-4 AttFPN map. Each stage has its own mask head producing 28×28 logits for 15 classes. One shared refiner upsamples these logits to 56^2, selects the most uncertain points on the current grid (196 per ROI in training, up to 784 at inference), refines them, and repeats at 112^2. Each point becomes a token

$$z_i = \mathrm{LN}(E_f(f_i) + E_m(m_i) + E_p(u_i) + e_r) \in \mathbb{R}^{128}, \quad (1)$$

where $f_i \in \mathbb{R}^{256}$ is the P2 feature, $m_i \in \mathbb{R}^{15}$ the original coarse logits, $u_i \in \mathbb{R}^2$ the ROI coordinate and e_r a level embedding. With Z the tokens of one region, the point logits are

$$\ell_i = W_o\, G(Z)_i + b_o \in \mathbb{R}^{15}, \quad (2)$$

where G is two pre-LayerNorm residual blocks and a final LayerNorm, using point-wise MLPs (MLP variant) or four-head self-attention within the region (transformer variant). The logits ℓ_i replace the selected grid values. Training adds a point loss to the detector loss:

$$L = L_{\text{base}} + \sum_{s=1}^{3} w_s\, \tfrac{1}{2}\big(L^{s,56}_{\text{point}} + L^{s,112}_{\text{point}}\big), \quad w = (1, 0.5, 0.25), \quad (3)$$

a binary cross-entropy on the ground-truth class channel against targets sampled from the ground-truth masks. The transformer was our proposed variant; the MLP was added as a matched control. They add 458,383 and 458,127 parameters respectively.

2 Setup

Official PIXray split (3,560 training and 1,486 test images, 15 classes), annotations unchanged. All methods train 60 epochs with AdamW (learning rate 10^{-4}, effective batch 16) and one seed; the final-epoch checkpoint is evaluated with COCO metrics. The refinement variants differ from the parent only in the ROI head. Baselines are adapted implementations under the same protocol.

3 Results

The MLP variant achieves the highest observed mask AP, 71.50%, adding about 0.46 million parameters (Table 1). It improves mask AP by +2.38 pp over the parent, +2.28 pp over the bilayer baseline, +10.08 pp over Occ and +2.51 pp over the PIDray adaptation, and also leads mask AP_{75} and mask AR@100. The gain grows from +0.39 pp at IoU 0.50 to +5.00 pp at IoU 0.95, so it comes from tighter masks; box AP does not improve (−0.21 pp). The transformer variant gains +0.95 pp; the MLP variant outperforms it by +1.43 pp, at all 10 IoU thresholds, in this run.

4 Error Analysis

On test instances matched by both methods, the MLP variant raises mask IoU by +1.07 pp and boundary IoU by +1.17 pp over the parent, improving 70.8% of them; Fig. 2 shows more complete masks from both of our variants than from any baseline. Refinement cannot fix missed or misclassified detections: the hardest categories (Dart, Saw_blade, Knife and Screwdriver) gain +2.96 pp, but Dart remains weakest at 36.87%. Point-to-point attention added no accuracy over the point-wise MLP in this single-seed comparison.

Table 1: PIXray official test split. Bold: best observed value. Δ: difference from the parent baseline in percentage points. One training seed. The bilayer baseline also uses a recall-enhanced RPN and auxiliary occluder masks; Occ uses SAM ViT-L and auxiliary occluder masks, with incompletely recorded prediction provenance. Baselines are adapted implementations.

Method	Params (M)	Box AP ↑	Mask AP ↑	Mask AP_{75} ↑	Mask AR@100 ↑	Δ Mask AP
Cascade Mask R-CNN + AttFPN	97.1	80.21	69.12	79.30	72.23	—
Cascade + Bilayer Mask Head	105.2	**80.23**	69.22	79.13	72.41	+0.10 pp ↑
Occ Adaptation	336.7	73.70	61.43	69.07	65.06	−7.70 pp ↓
PIDray R0/R1 Adaptation	98.7	80.12	68.99	79.14	72.14	−0.13 pp ↓
Point Refinement—Transformer (Ours)	97.6	80.01	70.07	78.62	73.16	+0.95 pp ↑
Point Refinement—MLP (Ours)	97.6	80.00	**71.50**	**81.41**	**74.42**	**+2.38 pp** ↑

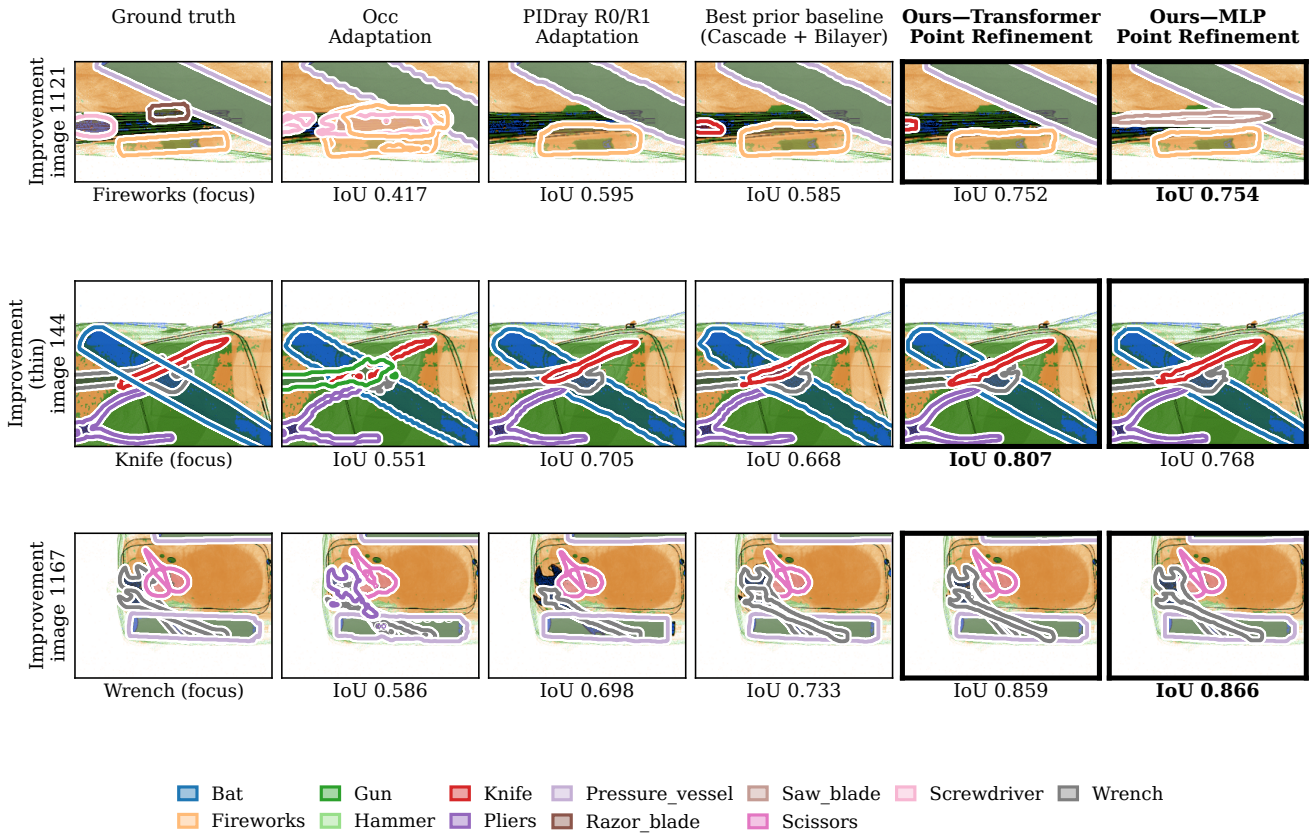


Figure 2: Illustrative improvement cases on identical crops, selected where both of our variants exceed all three baselines (row 2 from the hardest categories); not average performance. IoU is each method's best same-category mask IoU with the focus instance; every best mask is displayed (score ≥ 0.30).

5 Conclusion

Point-based refinement improves X-ray instance segmentation. Our MLP variant reaches the highest observed mask AP, 71.50% (+2.38 pp over its parent), with gains concentrated at strict IoU thresholds.